

Great job!!

COURSE: CHE 312: Transport Phenomena II

Spring Semester 2020

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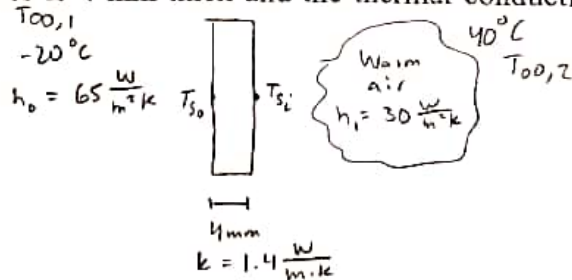
Quiz #1

February 5, 2020

Closed Notes and Book

Duration: 20 minutes

The rear window of an automobile is defogged by passing warm air at 40°C over its inner surface, and the associated convection coefficient is $30 \text{ W}/(\text{m}^2\cdot\text{K})$. Under conditions for which the outside ambient air temperature is -20°C and the associated convection coefficient is $65 \text{ W}/(\text{m}^2\cdot\text{K})$, what are the inner and outer surface temperatures of the window? The window glass is 4 mm thick and the thermal conductivity of glass is $1.4 \text{ W}/(\text{m}\cdot\text{K})$.



Assume steady state:

In - out + gen - cons = gaccum

In = out

$$\left(\frac{q}{A}\right) = h_o (T_{s,o} - T_{\infty,1}) = k \frac{T_{s,i} - T_{s,o}}{\Delta L} = h_i (T_{\infty,2} - T_{s,i})$$

$$q = \frac{T_{\infty,2} - T_{\infty,1}}{\sum \frac{1}{h_j A} + \sum \frac{L_i}{k_i A}}$$

$$T_{\infty,2} - T_{\infty,1}$$

$$\frac{q}{A} = \frac{1}{h_o} + \frac{1}{h_i} + \frac{\Delta L}{k}$$

$$\frac{q}{A} = \frac{(40 - -20)^\circ\text{C}}{\frac{1}{65 \frac{\text{W}}{\text{m}^2\cdot\text{K}}} + \frac{1}{30 \frac{\text{W}}{\text{m}^2\cdot\text{K}}} + \frac{4/1000 \text{ m}}{1.4 \frac{\text{W}}{\text{m}\cdot\text{K}}}}$$

$$\frac{q}{A} = h_o T_{s,o} - h_o T_{\infty,1}$$

$$T_{s,o} = \frac{\frac{q}{A} + h_o T_{\infty,1}}{h_o}$$

$$T_{s,i} = \frac{(\frac{q}{A} - h_i T_{\infty,2})}{h_i}$$

$\frac{q}{A}$ = # number
solve for temps.